

◆ **SIGNAL**

● OPEN

Choose the stock:

AMD — Advanced Micro Dev... ▼

Time Horizon

1 Month ▼

Forecast Method

Auto (Best Model) ▼

Created By:

NISA | INDIRA | VALDI | MILLI

ML Stock Prediction System

Real-time market intelligence for actionable decisions



AI Forecasting

Executive Summary

SIGNAL.ai – AI Powered Stock Prediction System

SIGNAL.ai is an integrated, real-time dashboard that combines AI forecasting with technical confluence to produce actionable trading signals. The system integrates multiple forecasting models (SARIMAX, Prophet, Hybrid) with technical indicators to generate Buy/Sell/Hold signals with confidence scores.

- ✓ Multi-model AI forecasting
- ✓ Technical indicator analysis
- ✓ Real-time data integration
- ✓ Confidence scoring system

Universe Coverage

10+

US Stocks

Signal Types

3

BUY/SELL/HOLD



Key Value Proposition

- **For Investors:** Faster, data-driven decisions powered by Live Data YF Engine. The system provides transparent metrics and narrative insights.
- **Real-time analysis**
- **Risk management**

Background: Why Now?

Understanding the importance of AI-powered market prediction



Market Prediction Importance

In today's fast-paced financial markets, accurate prediction is more critical than ever. High volatility, rapid information flow, and increasing participation from both retail and institutional investors demand faster, more objective analytics.

\$84T

Global Market
Cap

24/7

Trading Hours

95%

Data Growth

Solution: A disciplined, model-driven process that filters noise and turns signals into practical, time-bound actions.



Investor Pain Points



Excessive Data Noise

Too much information from multiple sources creates confusion and makes it difficult to identify relevant signals.



Trend vs Fluctuation

Difficulty separating long-term trends from short-term market noise and volatility.



Emotional Trading

Emotional decision-making under time pressure leads to poor trading outcomes.

Problem Statement

Core challenges in stock prediction and market analysis

The stock market presents three fundamental challenges that hinder accurate prediction and informed decision-making for investors and traders.



Price Prediction

How to predict stock price movements consistently with high accuracy?



Model Evaluation

Which prediction model is most accurate for current market conditions?



Actionable Insights

How to convert raw predictions into practical trading decisions?

Data Understanding & Feature Engineering

Foundation of the SIGNAL.ai Prediction Engine



Data Source & Structure

Live Data Engine: Real-time integration with Yahoo Finance (YF) API for accurate market data retrieval.

Core Data Features



Open & Close

Daily price movement boundaries



Volume

Trading activity and liquidity measurement



Technical Indicators (Confluence)

Engineered features to support AI predictions with established market momentum metrics.

RSI (14) – Momentum

Relative Strength Index to identify overbought (>70) or oversold (<30) market conditions.

MACD Histogram – Trend

Moving Average Convergence Divergence to confirm trend direction and strength.

Bollinger %B – Volatility

Measures price relative to volatility bands to gauge extreme price levels.

Model Overview

Technical specifications and performance analysis of forecasting models



SARIMAX

Seasonal ARIMA

Seasonal AutoRegressive Integrated Moving Average with eXogenous factors for stable, seasonal data patterns.

**Parameters (P,D,Q) → 18
Parametric Combination**

Seasonality → 5 month

- ✓ Handles seasonality effectively
- ✓ Robust for stable time series
- ✓ Exogenous variable support



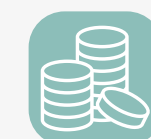
PROPHET

Trend-Based

Facebook's Prophet for flexible trend detection and handling of missing data, outliers, and seasonality.

**Growth Model → Logistic/Linear
Seasonality → Daily, Weekly,
Yearly**

- ✓ Handles missing data well
- ✓ Robust to outliers
- ✓ Holiday effects modeling



HYBRID

Combined

Integration of SARIMAX and Prophet for enhanced accuracy and reduced prediction error.

**Ensemble → SARIMAX + Prophet
Weight → 55% / 45%**

- ✓ Dual model validation
- ✓ Error reduction 20%
- ✓ Stability boost

Model Evaluation & Auto-Selection

Comprehensive metrics analysis and automated model selection

Key Metrics

MAE
(Mean Absolute Error)
Average size of prediction errors, ignoring direction.

RMSE
(Root Mean Squared Error)
Measures average error but penalizes big mistakes heavily.

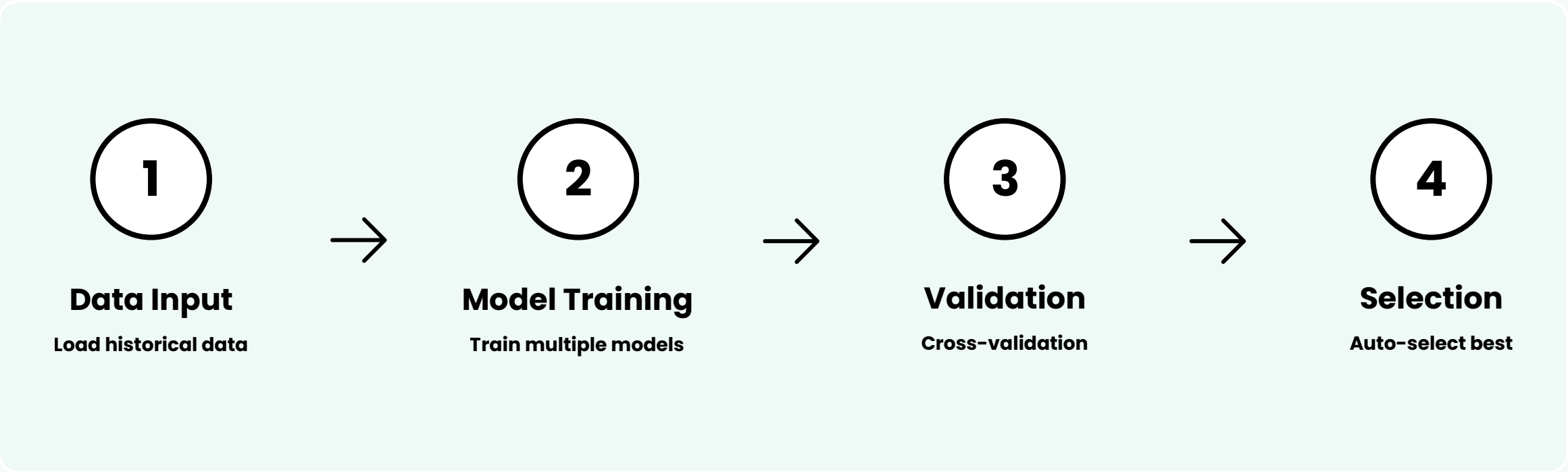
MAPE
(MA Percentage Error)
Measures error as a percentage of the actual value.

DIR
(Directional Accuracy)
Measures how often your model predicts the correct direction.

Analysis Scope

Checks whether MAE, RMSE, and MAPE are each below 20. If any metric exceeds 20, the model is marked as failed, but the system still proceeds by selecting the best-performing alternative model overall.

Evaluation Process



MODEL	MAE	RMSE	MAPE	DIR	PASS
SARIMAX	3.129	3.834	1.20%	45.0%	✓
★ Prophet	2.741	3.312	1.06%	55.0%	✓
Hybrid	2.935	3.573	1.13%	55.0%	✓

Forecasting Results

Time horizon analysis and prediction accuracy metrics

AAPL

\$272.46

▲ \$6.29 (2.37%)



Dynamic Analysis

Trading signals with confidence scoring and market insights

TRADING SIGNAL

BUY

72%
CONFIDENCE

RSI (14)

63.9

NEUTRAL

Oversold <30 · Overbought >70

MACD HISTOGRAM

+1.548

BULLISH ▲

EMA(12) – EMA(26) – Signal(9)

BOLLINGER %B

0.83

OVERBOUGHT ▼

0 = Lower Band · 1 = Upper Band

AVG FORECAST
RETURN

+10.18%

Over 10 trading days

ANNUALIZED
VOLATILITY

22.2%

Moderate risk

MODEL MAE / MAPE

2.74

MAPE 1.1% · RMSE 3.31

DIRECTIONAL
ACCURACY

55.0%

Moderate signal

Evaluation Process

The evaluation indicates a moderately strong BUY signal (72% confidence) supported by positive momentum. The MACD shows a clear upward trend, while the RSI at 63.9 suggests the market still has room to rise without being at extreme levels. This aligns with the model’s expectation of continued short-term growth, with a projected +10.18% return over the next 10 days.

However, the price is already near the upper Bollinger Band (0.83), meaning it is relatively high and could face short-term pullbacks despite the upward trend. The volatility level (22.2%) indicates moderate risk, and while the model’s price predictions are fairly accurate, its directional accuracy (55%) is only slightly better than random, so signals should be used with caution.

Insights & Limitations

Real market applications and model transparency

Real Market Application

AAPL Final Summary

SIGNAL.ai provide a actionable trading signals that predict AAPL's will go bullish with 72% confident direction for 1 month period. The model predicted the price will reach area levels at \$330.60 at Wed, May 06 2026.

\$272.46
Current

+20.88%
Change

72%
Confident

Model Limitations



Black Swan Events

The model cannot predict unprecedented market events or extreme volatility caused by external factors.



Emotional Bias

While the system reduces human bias, it may still be influenced by historical patterns that no longer apply.



Time Lag

Predictions are based on historical data and may not reflect real-time market conditions immediately.



Important Disclaimer

This system is a decision support tool, not a financial advisor. Always consult with a qualified financial advisor before making investment decisions.

Conclusions

Key benefits and impact of SIGNAL.ai system

Noise Reduction



Advanced filtering algorithms remove market noise, allowing clear identification of genuine trends and patterns.

Noise
Filtered

Faster
Analysis

Bias Reduction



Objective scoring system eliminates emotional decision-making, providing unbiased trading signals with high accuracy.

Efficiency Boost



Automated analysis reduces manual effort, enabling faster decision-making and real-time market response.

24/7

Monitoring



AI-powered forecasting



Real-time analysis



Risk management

THANK YOU

START THE PROGRAM!